Dissertation on

Context Extraction from Videos

Submitted in partial fulfilment of the requirements for the award of degree of

Bachelor of Technology
in
Computer Science & Engineering

UE18CS390B – Capstone Project Phase - 2

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CERTIFICATE

This is to certify that the dissertation entitled

Context Extraction from Videos

is a bonafide work carried out by

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In partial fulfilment for the completion of seventh semester Capstone Project Phase - 2 (UE18CS390B) in the Program of Study -Bachelor of Technology in Computer Science and Engineering under rules and regulations of PES University, Bengaluru during the period June 2021 – Nov. 2021. It is certified that all corrections / suggestions indicated for internal assessment have been incorporated in the report. The dissertation has been approved as it satisfies the 7th semester academic requirements in respect of project work.

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DECLARATION

We hereby declare that the Capstone Project Phase - 2 entitled **"Context Extraction from Videos"** has been carried out by us under the guidance of Prof. Jeny Jijo, Assistant Professor and submitted in partial fulfilment of the course requirements for the award of degree of **Bachelor of Technology** in **Computer Science and Engineering** of **PES University, Bengaluru** during the academic semester June – Nov. 2021. The matter embodied in this report has not been submitted to any other university or institution for the award of any degree.

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ABSTRACT

Text summarization and organization of information in the educational domain. Automating the process of making notes could be pitched as a way to alleviate students' concern that their own notes won't capture everything. These concerns have been solved in the past using a video summarizer or a transcript generator like VoicesEvas. A potential solution to this problem is addressed by compacting a video to its most important components in the form of text. Automatic summarizations can generate semantic descriptions to lecture videos, which can then be used for indexing topics and facilitating review. State of the art algorithms are used for all three stages i.e. transcript generation, text summarisation and topic labelling.

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CHAPTER-1
INTRODUCTION

The 21st century has brought about a massive change in the world of education. The era of offline classes with teachers and students within the confines of a classroom is slowly coming to an end. The internet deserves massive credit in its role to effectively shift the paradigm in the way in which learning is done. It has effectively taken learning beyond the sacred walls of schools, universities, educational institutions and directly into the people's hands. The recent COVID-19 pandemic further illustrates the importance of online learning in today's education systems, as it has proven to be a blessing in disguise to both students and teachers alike who are unable to attend school due to the risk of the virus being spread. Beyond high school, online learning and education is steadily increasing its market share at the pre-university level. Furthermore, e-learning is expanding its presence beyond the traditional fields as well. Unsurprisingly, there is an increasing demand for highly optimized learning applications in the field of online education as a whole.

Thus, we arrive at our project, which aims to address one of the challenges faced by students everywhere which involves creating appropriate summaries of their class lecture videos in order to maintain a useful set of short notes for all the lessons and automatically generate labels for all the lecture videos which could better aid them in their endeavors.

The goal of our product is to provide students with a concise summary of lecture videos by making use of concepts in text summarizations and to appropriately extract the relevant topics from the video using topic modelling in order to aid in their classifications overall.

One of the benefits involved with automating the process of summarizing lecture videos is that it saves students a lot of valuable time which they could spend on other important activities. Our product can also aid in the organization of lecture videos by assigning accurate labels to them which could be used in other applications that appropriately store them.

The system being proposed is going to be designed and implemented from scratch with no foundation on any prior systems.

CHAPTER-2

PROBLEM STATEMENT

This project is mainly focussed on providing innovative solutions to challenges faced by students in the online educational domain and also providing crucial insight into this ever growing technological domain which will transform our lives in the future.

Our problem statement can be compartmentalised into three major phases:-

1) Extracting audio transcripts from the videos.

2) Applying text summarization to get a brief description of the lecture videos.

3) Identifying topics consisting of relevant labels for all transcripts within the dataset.

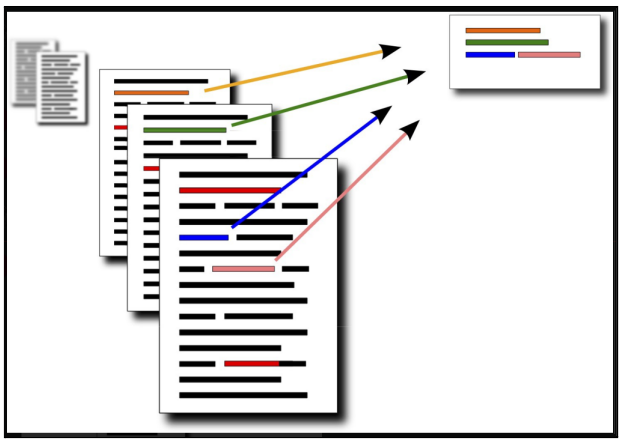

Fig 2.1: An illustration of topical document clustering

CHAPTER-3

LITERATURE REVIEW

3.1 Domain taxonomy learning from text: The subsumption method versus hierarchical clustering

3.1.1 Introduction

The authors found that "the optimized settings in the term extraction step require that the domain pertinence filter has a higher weight than the other filters. The optimized settings for the subsumption method depend on the user's preferences regarding the depth and quality of the relations in the taxonomy. A taxonomy with greater depth has more information about the taxonomic relations, but these taxonomic relations are less reliable. A decision support criterion based on taxonomy quality and average depth that considers the user-preferred settings"[1] was put forward. "For hierarchical clustering, complete linkage provides better results than single and average linkage. The best-performing distance measure is the window co-occurrence similarity with a window size of 11."[1] The authors "compared the subsumption method to the hierarchical clustering algorithm. Users who are primarily interested in the quality of the taxonomic relations should apply the subsumption algorithm with a high threshold value. Users interested in a deep taxonomy should apply the hierarchical clustering algorithm with the previously determined optimized settings."[1]

3.1.2 Characteristics and Implementation

The ADTCT framework breaks down the task of Taxonomy Learning into 3 major steps. The first step involves extracting the terms from the text corpora. A part of speech parser is used to process the sentence into a tree-like structure, where the leaves of the tree are essentially the nouns and all internal nodes are the tags. The next step is to apply Term Filtering. "*Terms are selected based on domain pertinence, lexical cohesion, domain consensus & structural relevance*."[1] The adapted formulae used to calculate these scores have been presented in the paper. The last and final step is

the concept hierarchy creation. Two methods are discussed and distinguished to determine the relations. "*They are the Subsumption Algorithm and Agglomerative Hierarchical Clustering respectively. The Subsumption method uses the concept of co-occurrence. Concept co-occurrences are determined using which potential parents/subsumers are identified.*"[1] The clustering algorithm works similar to the traditional algorithm except that the 'document co-occurrence' & 'window-based' similarities are used as the distance measures for grouping the clusters together.

3.1.3 Model

ADTCT framework - The framework built by authors who proposed the paper

3.1.4 Evaluation

The evaluation framework presented by the authors, compares the automatically created taxonomy created by the ADTCT framework with "a golden taxonomy created by experts from economics and management. Several measurements are used to conduct the evaluation. Lexical Precision is used to analyze relevance of found terms. Common semantic copy of a node is used to determine the quality of relation between a concept and its sub / super concepts in both the taxonomies. Using the local measurements, two global measurements, TP and TR which stand for Taxonomic Precision & Taxonomic Recall are used to find the overall F-score." [1]

3.1.5 Performance

The TF values of the taxonomy constructed by ADCTC were better when compared to the original taxonomy for a range of values of t. The average depths of the taxonomy constructed were smaller than the original taxonomy for a range of values of t.

3.2 Automatic Text Summarization: A State-of-the-Art Review

3.2.1 Introduction

This paper elaborates on the shift from extractive to automatic summarization in the last years. Despite the emergence of abstractive summarization techniques many challenges still remain unresolved.

3.2.2 Characteristics and Implementation

"A fully data-driven approach to abstractive sentence summarization is introduced. Their method utilizes a neural **attention-based model** that generates each word of the summary conditioned on the input sentence."[2] "They combine this probabilistic model with a generation algorithm which produces accurate abstractive summaries."[2]

The model later extended to **RNN** which outperformed on the corpus used. In the switching **Generator/Pointer model**, the decoder is equipped with a "switch" that decides between generating a word based on the context or using a word from the input. The **"hybrid pointer-to-pointer generator network and the coverage mechanism"[2]** "uses pointing to copy words from the input text which provides to better accuracy and is better in dealing with OOV words, while retaining the ability to generate words."[2] "Sentence compression techniques for **query focused summarization**."[2] Sentence Ranking-Support Vector Regression, LambdaMART. Sentence Compression- rule-based, sequence-based and tree-based and Post-processing-coreference resolution and sentence ordering are performed. Additionally, "query attention model which learns to concentrate on different parts of the query at different points in time"[2] and a "diversity based attention model which aims to mitigate the drawback of generating repeated phrases in the summary"[2] were included.

Update Summarization - DualSum mode is "a variant of Latent Dirichlet Allocation (LDA) and learns to distinguish between common information and novel information."[2] ILP model used in update summarisation as it uses a "set of rich features to measure the importance and novelty of

the bigram"[2] and "design a sentence reranking component which allows to explicitly model a sentence's importance and novelty."[2]

Personalization - Using explicit (e.x. asking user) or implicit methods(e.x. observing user behaviour)

3.2.3 Model

"Extractive summarization as a sequence labelling task"[2] A single document summarization framework based on hierarchical document encoder and an attention-based extractor. RNN based sequence model for extractive single-document summarization

- Abstractive methods - Encoder-Decoder Model,RNN,Switching Generator/Pointer model. Textsum by Google.
- Neural summarization technologies for articles containing thousands of words.
- A hybrid pointer-to-pointer generator network and the coverage mechanism. **Abstractive query summarization .**
- **Update Summarization**- Supervised Integer Linear Programming (ILP) framework.
- **Personalization**

3.2.4 Evaluation

Evaluation can be in terms of intrinsic (tests the quality of summarization itself, e.g. its informativeness and coherence) or extrinsic evaluation(checks the effectiveness of a summary for another task, for example, answering a query). Two manual methods are pyramid and responsiveness -Subjective and difficult to reproduce. Automatic summarization evaluation measure is ROUGE.

3.2.5 Performance

ROGUE is used for evaluation. Rouge for abstractive summaries outperforms the extractive summaries For query based system yields 11.02 ROUGE-2 score. For query-based summarization, rouge yields 28% in ROUGE score.

3.3 Semantic Text Summarization of Long Videos

3.3.1 Introduction

"This paper introduces a novel method for both video summarization and annotation. Frame to frame motion, frame image quality, as well cinematographic and consumer preference are uniquely fused together to determine interesting segments from long videos. Key frames from the most impactful segments are converted to textual annotations using an encoder-decoder recurrent neural network. Textual annotations are summarized using extractive methods where LSA, LexRank and SumBasic approaches performed best."[3]

3.3.2 Characteristics and Implementation

"The sumy 0.4.1 python framework along with NLTK libraries were used to evaluate Luhn's algorithm, Edmundson's heuristic method, Latent Semantic Analysis (LSA), LexRank, TextRank, SumBasic and KL-Sum text summarization techniques.Video Captioning: Modifying S2VT with new frame features and introduction of key frame selection. Each keyframe is passed through the 152-layer ResNet CNN model."[3]

3.3.3 Model

Videoclip captioning: 152 layer ResNet CNN model.

They have used two layered LSTM fixed to 80 time steps.

3.3.4 Evaluation

Text Summarization: ROUGE Superframe cut selection: An ablation analysis across the six features of Boundary, Attention, Contrast, Sharpness, Saturation, and Face impact was performed across all 25 videos

3.3.5 Performance

"Superframe cut selection: For the SumMe dataset, the six features had an overall RMSE of 0.0271 as compared to the ground truth, showing this suite of features are excellent indicators of frame relevance."[3]

3.4 Multi-Document Topic Segmentation using Bayesian Estimation

3.4.1 Introduction

This paper presents a "novel approach to document segmentation. Our proposed technique leverages text similarities so that accurate word distributions associated with text segments can be computed. This is done by using other documents which provide counts of relevant words. These counts come from sentences that are similar to the ones in the target document. By incorporating this information, we address one of the problems with Bayesian segmentation: estimation of segment likelihood using scarce data. The proposed technique was evaluated on a multi-document scenario where different learning materials from different media needed to be segmented. Studies show that this technique effectively obtains better document segmentation, outperforming the baseline (which corresponds to a state of the art technique)."[4]

3.4.2 Characteristics and Implementation

The BayesSeg approach proposed by the authors "allows us to incorporate external information more easily than other approaches. Another important advantage is that it does not need a prespecified number of topics or segments, which would be difficult when dealing with multiple documents."[4] "This technique translates lexical cohesion into a Bayesian setting by computing compact and consistent lexical distributions. This is done by modelling words in the segments as draws from multinomial language models. This means that if a sentence t is in segment s, the corresponding bag-of-words xt is considered as having been drawn from language model theta."[4] The observation likelihood can be calculated using the above parameters. DCM is used to avoid searching the full space of the language models. Using a matrix, the DCM distribution is being fitted to all segmentation probabilities. The values of the matrix show likelihoods of individual segments. "Using DP, the best segmentation up to a sentence can be iteratively computed. This will finally help in finding the overall likelihood of the full segmentation of the document. The above traditional method has the problem of computing word distributions using scarce data, since segments generally do not contain many sentences."[4] To overcome this information from other sources are also considered to provide accurate models.

3.4.3 Model

The authors have proposed the use of an approach called the BayesSeg approach and have enhanced it by considering information from other sources to improve the accuracy. They call the enhanced version, BayesSeg MD.

3.4.4 Evaluation

The unsupervised segmentation approaches, BayesSeg and BayesSeg MD, make use of evaluation metrics Pk and WindowDIff. "Both metrics slide a window through the document. The Pk metric assesses whether the sentences on the edge of the window belong to the same segment or not,

when compared to reference segmentation. The WindowDiff is stricter as it counts the total number of segment boundaries in the full length of the window. ”[4]

3.4.5 Performance

8 of the total 10 test cases showed that “BayesSeg MD outperformed BayesSeg”[4]. From the results it was possible to say “that it is possible to achieve more accurate statistical models by incorporating information from other multimedia sources.”[4]

Further Literature Survey:

A detailed literature survey of several papers is provided in the link below.

https://drive.google.com/file/d/1PGN3N6t6RiSfOy5Yj_-_0edV-oVt6ZRN/view?usp=sharing

CHAPTER-4

DATA

4.1 Overview

Math Lectures: Subtitles from YouTube videos

The dataset in particular was compiled by scraping through educational lecture videos consisting of various concepts in the fields of Computer Science and Mathematics.

The dataset consists of two columns:

text - Speech transcripts from the video.

label - A generalised text label designated to each transcript that references the underlying topic in the lecture video.

4.2 Dataset

https://www.kaggle.com/extralime/math-lectures

	text	label
0	The following content is\nprovided under a Cre...	Calculus
1	In this sequence of segments,\nwe review some ...	Probability
2	The following content is\nprovided under a Cre...	CS
3	The following\ncontent is provided under a Cre...	Algorithms
4	The following\ncontent is provided under a Cre...	Algorithms
...	...	...
855	The following content is\nprovided under a Cre...	Math for Eng.
856	>> [MUSIC] >> DAVID J. MALAN: All ...	Diff. Eq.
857	The following content is\nprovided by MIT Open...	Linear Algebra
858	The following content is\nprovided under a Cre...	AI
859	The following content is\nprovided under a Cre...	Diff. Eq.

860 rows × 2 columns

Fig 4.2.1: Sample Dataset

CHAPTER-5

PROJECT REQUIREMENTS SPECIFICATION

5.1 Product Features

There are two main features of our product:-

a) A Brief Summary of the Lecture Videos - The first feature of our product is to provide students and teachers with a brief description of every lecture. The audio components of the lectures will be extracted & existing signal processing techniques will be used for text conversion. State of the art extractive summarization techniques will be used to compose a condensed, readable and concise synopsis. This would not only help students, but also teachers in their endeavours.

b) Topic Modelling - The second and most significant feature of our product involves formulation of labels using the latest machine learning techniques currently in use. Topics which essentially consist of weighted clusters of keywords are extracted from the lecture transcripts.

5.2 User Classes and Characteristics

Our product's application is generally in the educational domain and could potentially revolve around two fundamental classes, namely students and teachers. Both classes are provided with the same level of access to our product functions

5.3 Operating Environment

An ideal environment for the implementation of our project would be a laptop/desktop with 8GB of RAM with an i5 processor or higher, with Windows 10 being the operating system which is used.

5.4 General Constraints, Assumptions and Dependencies

Important constraints could include but are not limited to :-

- The implementation of advanced document clustering techniques.
- Difficulties in extracting the speech transcripts from the lectures due to possible disruptions in sound quality.
- The possibility of the model overlooking certain important keywords from the transcripts during text summarization.
- There's also the possibility of the model overemphasizing certain words which might be trivial.
- Providing an accurate topic name which would encompass all the relevant videos it covers.

Our current application for the capstone project is being tailored towards a specific educational lecture video dataset obtained by compiling transcripts from around 830 educational lecture videos focussed on computer science and mathematics in general. Future versions could be trained on a variety of educational datasets to the point of it being fully functional on any given input, at which point it could become commercially viable.

5.5 Risks

The application could fail due to inadequate resource requirements or unforeseen errors in the code.

5.6 Functional Requirements

- Transcript generation : The complete lecture video is initially taken as the raw input. The transcripts from the lectures are extracted using IBM Watson, which is an advanced open source speech to text API making use of a voice recognition library.

- Text Summarization: The user can choose between two options depending on what they need. The first feature is a text summarization. The user has a choice of uploading a new video or directly uploading a lecture video transcript. If the user uploads the wrong type of file, the system will not accept the input.

- Topic modelling: The second option is topic modelling. When a user chooses topic modelling, the system generates topics and displays it on the user interface. Each topic consists of a collection of weighted keywords which are also displayed respectively.

5.7 External Interface Requirements

5.7.1 User Interfaces

Various user interfaces for the product could be:-

1) Interface for the video file is uploaded/given as input.

2) The backend will generate transcripts for the videos and display the transcript.

3) Using the transcript, the text to be summarized will be sent to the server and the resulting package will be the summary of the text. The interface of the printed summary will be shown.

4) The server generates topic labels for the transcript and sends it to the user interface.

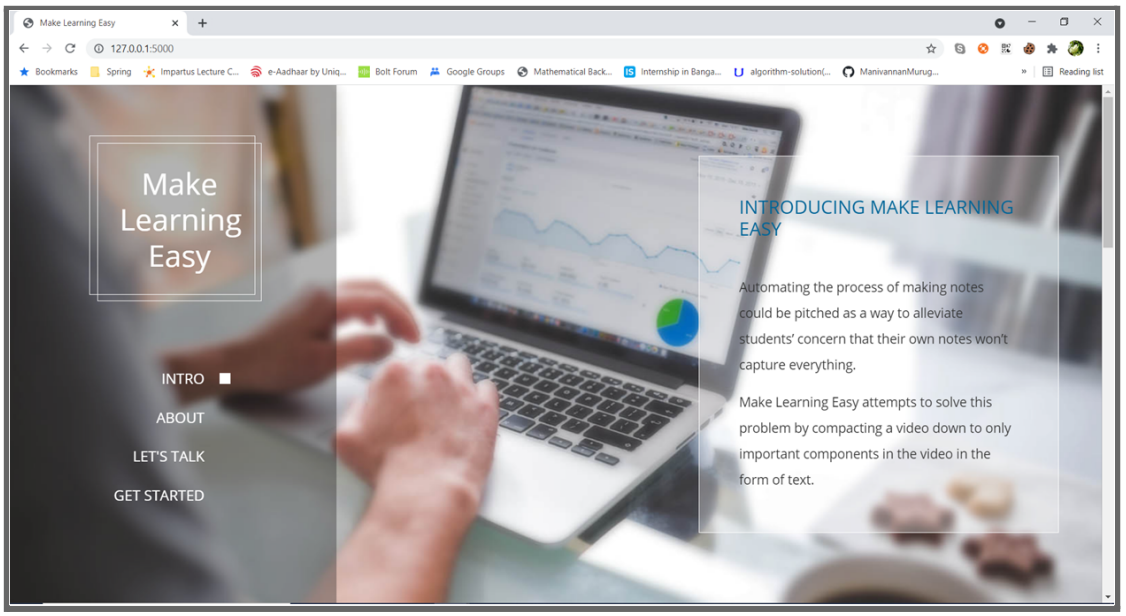

Fig 5.7.1: Prototype interface of the browser

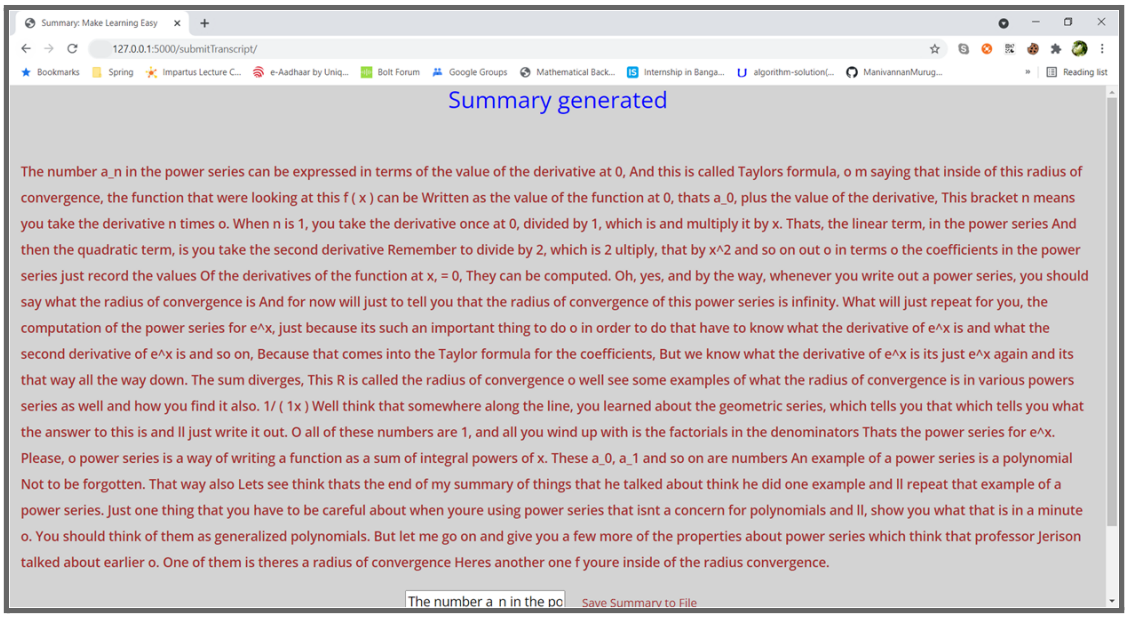

Fig 5.7.2: Prototype Interface of Summary

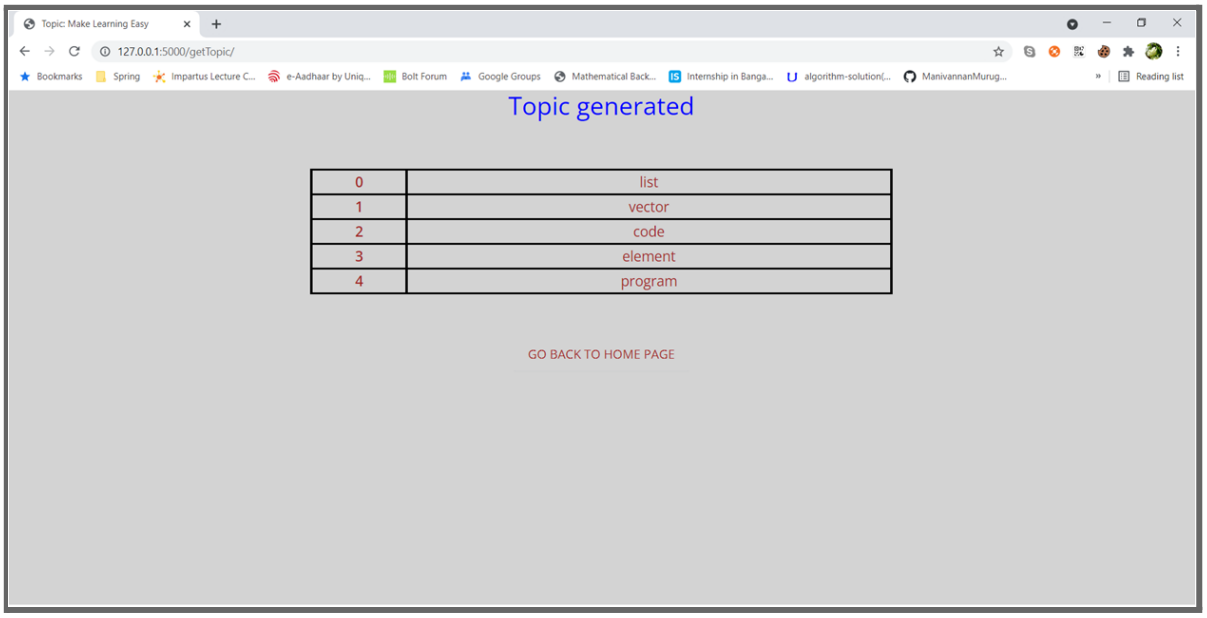

Fig 5.7.3: Prototype Interface of the topic (keywords) generated

5.7.2 Hardware Requirements

The system must run over the internet, all the hardware shall require connection to the internet and will be a hardware interface for the system. As for e.g. Modem, WAN – LAN, Ethernet Cross-Cable. The backend must have a smart CPU and a powerful GPU. The high GPU requirement is to train the model faster to increase performance.

The user interface for the software shall be compatible with browsers such as Internet Explorer, Mozilla by which users can access the system. The web pages can communicate via HTTP protocol standards.

5.7.3 Software Requirements

Home Page: Introduction to what the website is about. Links are also present to navigate to other web pages.

Version Number-2

Operating Systems Supported -Windows and other standard operating systems

Tools and Libraries- Python libraries like Flask, requests to get the url and perform manipulations.

Upload Video/ Choose from existing video: User is given an option to upload a new video or select an existing transcript.

Version Number-2

Operating Systems Supported - Windows and other standard operating systems

Tools and Libraries- Requests, IBM Watson, Punctuator.

Text Summarization: Printed summary of the video uploaded will be displayed after contacting the client.

Version Number-2

Operating Systems Supported - Windows and other standard operating systems

Tools and Libraries- TextRank is used to generate summaries.

Topic Labels: Get topics for each summarized text.

Version Number-2

Operating Systems Supported - Windows and other standard operating systems

Tools and Libraries- Gensim, PyLDAVis, Numpy and Pandas

5.8 Non-Functional Requirements

5.8.1 Performance Requirement

Since our product heavily relies on the quality of the audio transcripts, minimizing the disruptions & disturbances taking place in audio signals could be looked at for future prototypes. Domain pertinence is also another such aspect which could be improved upon. Further optimizations on time consuming components such as the speech to text and text summarization modules could effectively enhance the throughput of the application.

5.8.2 Safety Requirements

The first product feature which has been specified, ensures that a brief and concise description of the lecture is provided. Safety measures have to be taken when it comes to generating these summaries. It is highly essential for the product to ensure that no personal details by any means are specified in the summaries which are being generated. Since this might lead to a lot of privacy issues, appropriate safety measures must be taken to ensure the privacy of the students as well as the teachers. The same aspect holds good while creating the ontology of the concepts which are found by the model. Security Requirements we identified that the main users of our product would be students, teachers, lecturers and other staff pertaining to academics. When it comes to accessing lecture videos, users have to be authorized by the university/institution and must have the required permissions to view these videos. In the same manner, the summary and ontology generated by our model must also be accessible only by the users who have been permitted by the respective institution. If the product is going to be integrated with the school's official website, the identity authentication utilized by the website would suffice. If the product is going to be stand alone, users will be authenticated by the product’s own confirmation feature.

5.9 Other Requirements

Lectures which are taken in universities are generally taken on a day-to-day basis. This requires a lot of storage since every lecture would at least be for an hour and having plenty of such lectures across all the semesters would definitely need huge space requirements. Maintenance of these lecture videos would essentially mean ensuring that all the lectures are up to date. This would help in preventing any redundant lectures which have been taken in the past.

CHAPTER-6

SYSTEM DESIGN AND PROPOSED METHODOLOGY

6.1 Introduction

Our project aims to address one of the challenges faced by students everywhere which involves creating appropriate summaries of their class lecture videos in order to maintain a useful set of short notes for all the lessons and automatically generate labels for all the lecture videos which could better aid them in their endeavors.

6.2 Current System

The system being proposed is going to be designed and implemented from scratch with no foundation on any prior systems.

6.3 Design Considerations

6.3.1 Design Goals

- **Our proposed system is as follows:**

  We use one of the services of **IBM Watson** for generation of transcripts. "IBM Watson Studio is an integrated environment designed to develop, train, manage models, and deploy AI-powered applications and is a Software as a Service (SaaS) solution delivered on the IBM Cloud. The IBM Cloud provides lots of services like Speech To Text, Text To Speech, Visual Recognition, Natural Language Classifier, Language Translator, etc. The Speech to Text service transcribes audio to text to enable speech transcription capabilities for applications.."[15]

- In order to facilitate the process of text summarization, we make use of TextRank which is an extractive and unsupervised text summarization technique. TextRank is a text summarization technique which is used in Natural Language Processing to generate Document Summaries. This module gives the user a short, concise and brief summary of the video lecture that has been submitted by the userIt uses an extractive approach and is an unsupervised graph-based text summarization technique.

  The original document is initially split into individual sentences systematically. The sentences are then preprocessed by removing unnecessary stop words & are later converted to their corresponding vector representations. Once the vector representations of all the sentences within a document are obtained, weights are appropriately assigned to the edges between each vertex, with each vertex essentially representing a particular sentence. Once the weights, edges and vertices are calculated, using the networkx library, a textrank graph can be constructed. Based on the importance of the sentences within the graph, the top N sentences can be selected. This is extremely similar to the way pagerank is used to measure the importance of webpages on the internet.

- Some of the fundamental pre-processing steps taken to refine the transcripts for topic modelling are :-
  - ➔ *Tokenization*: Split the text into sentences and the sentences into words.
  - ➔ Lowercase the words and remove punctuation.
  - ➔ Words that have fewer than 3 characters are removed.
  - ➔ All stopwords are removed.
  - ➔ Words are lemmatized — words in third person are changed to first person and verbs in past and future tenses are changed into present.
  - ➔ Words are stemmed — words are reduced to their root form.
  - ➔ *Token filtration*: Finally, only the tokens which appear in more than 15 transcripts are taken into consideration.

- For the topic modelling, we have incorporated the LDA model using the Bag of Words approach and the TF-IDF approach. Topic modeling is a type of statistical modeling for discovering the abstract "topics" that occur in a collection of documents. "Latent Dirichlet Allocation (LDA) is a popular topic modeling technique to extract topics from a given corpus. The term latent conveys something that exists but is not yet developed. In other words, latent means hidden or concealed. It classifies or categorizes the text into a document and the words per topic, these are modeled based on the Dirichlet distributions and processes."[16] The model is trained using the dataset comprising 860 lecture video transcripts. Using the LDA model, N number of topics can be obtained, where each topic is a collection of keywords along with corresponding weights. The video is initially associated with the most relevant topic by score and based on the topic, the tags are obtained. Using this model, an unseen lecture video transcript can be fed into our system & corresponding tags can be procured.

  The LDA makes two important assumptions:

  1. Documents are a mixture of topics
  2. Topics are a mixture of labels

  "And, these topics using the probability distribution generate the words. In statistical language, the documents are known as the probability density (or distribution) of topics and the topics are the probability density (or distribution) of words."[16]

- We use hyperparameter tuning i.e. we use the coherence score to find the optimal number of topics to be set initially for the LDA model to work best.

- The optimal number of topics for the BOW model is around 32.

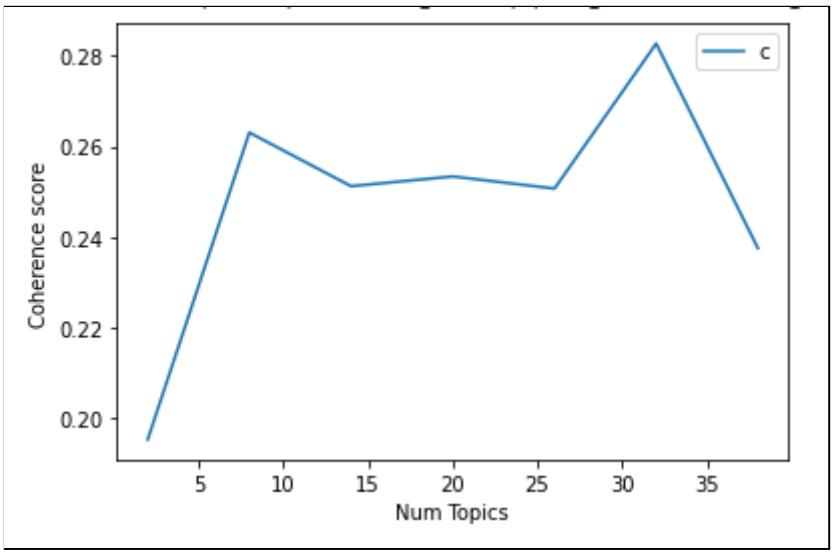

Fig 6.3.1: Coherence score for the LDA BOW Model

- The optimal number of topics for the TF-IDF model is around 12.

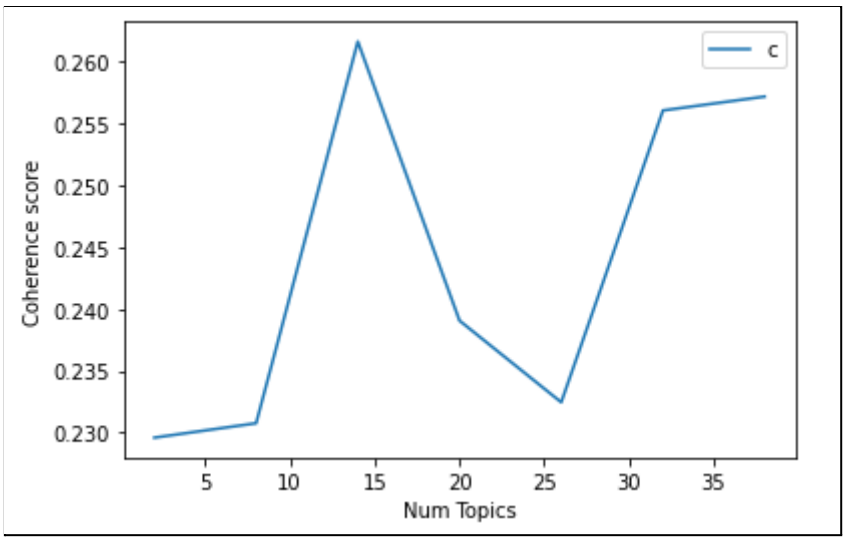

Fig 6.3.2: Coherence score for the LDA TF-IDF Model

Topic Modeling Visualisation:

The thirty most relevant terms for all the topics are plotted. When one of the topics is clicked (say 5), the thirty most relevant terms for that topic (topic 5) is displayed.

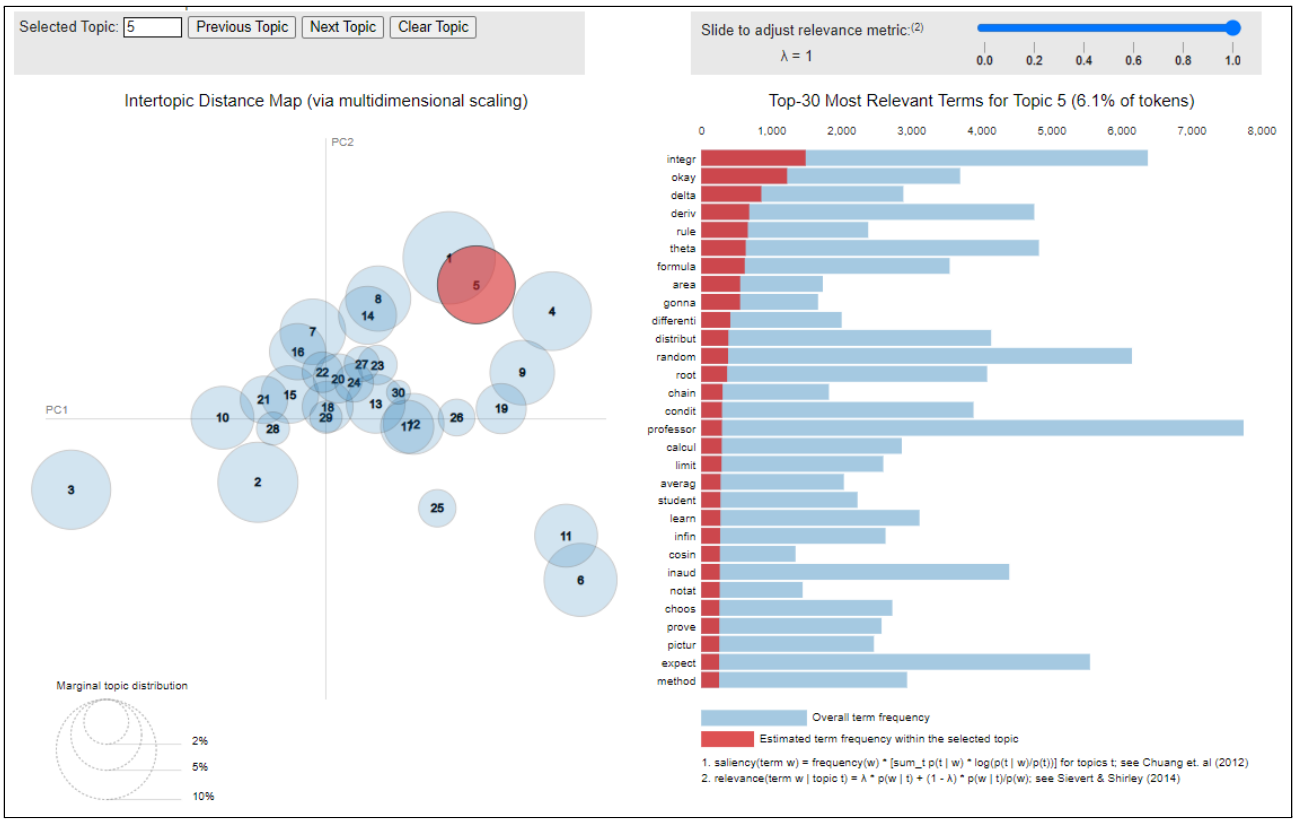

Fig 6.3.3: 30 most relevant topics for the BOW LDA model

Similarly, the thirty most relevant topics can be obtained for the TF-IDF model.

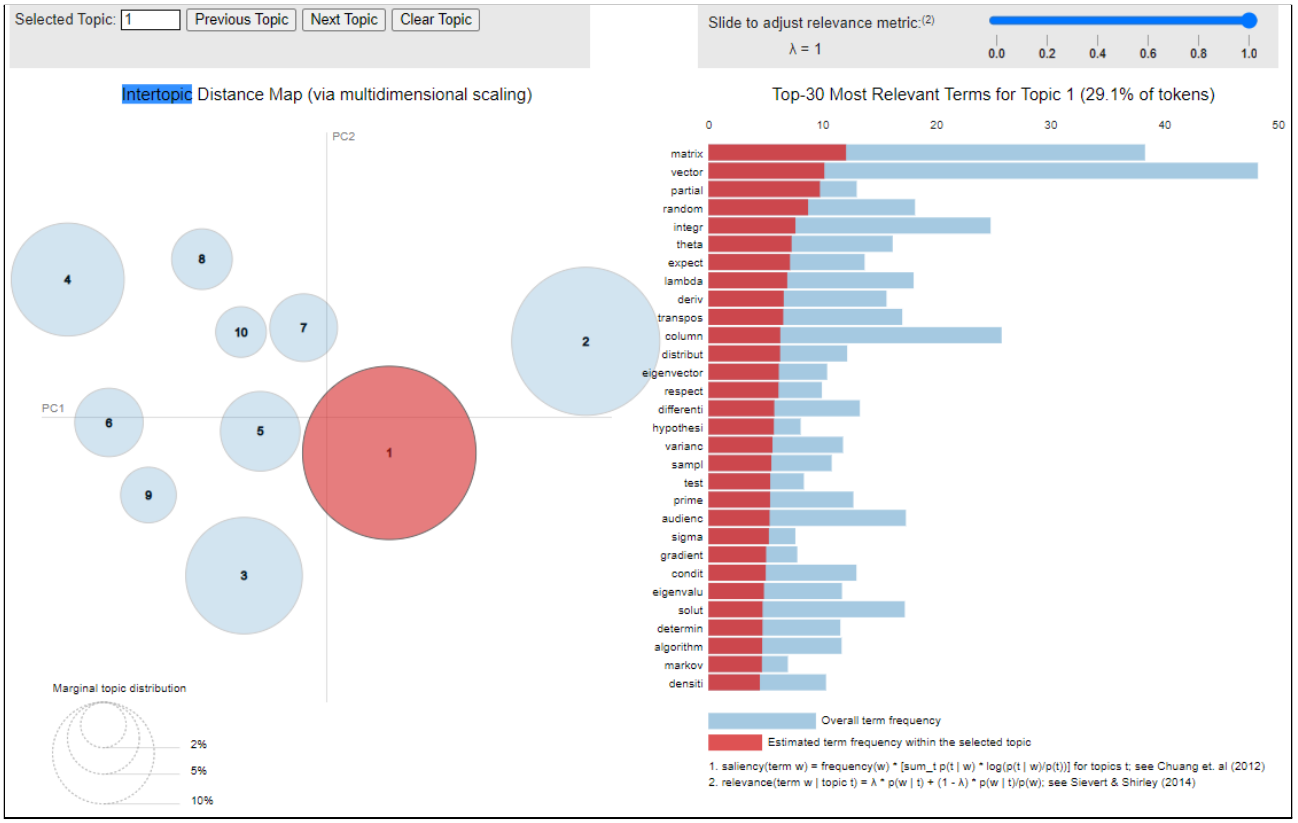

Fig 6.3.4: 30 most relevant topics for the TF-IDF LDA model

6.3.2 Architecture Choices

The architecture we considered for the design is the requirement gathering, work log recording and design decisions, valid business case, document evidence of requirement management, conceptual architecture, requirements realization views, business, data, application and technology views, the roadmap and implementation views and layered views.

The breakdown of the different levels and design gives a clear understanding and systematic approach towards development and deployment of the final product. The disadvantage can be the challenges faced in the modification of the design and updating all the document specifications accordingly.

6.3.3 Constraints, Assumptions and Dependencies

- The **assumptions** we make is the selection of various approaches for implementation based on the definition and domain of our dataset.
- The **dependencies** for topic modeling and text summarisation are the transcripts of the lecture videos.
- The system requires a high speed GPU so that the system is fast and the **performance** is high.
- Since our product heavily relies on the quality of the audio transcripts, it is extremely important to ensure that we can minimize the disruptions & disturbances taking place in the audio signals. Although it is impossible to completely remove the **quality** issues, efforts have to be taken to reduce them.
- Another important aspect determining the performance of our product would be the textual transcripts themselves. Since the summaries which would be generated would rely on the conversation happening in class, there is a high chance that the discussion might steer its way to being something very trivial, for example the teacher might be talking to a student

about something which is not relevant to the subject, in this case the model has to ignore these textual transcripts.

- There is also a chance that the model might not give importance to words which are relevant to the domain. Hence **domain pertinence** must also be given at most importance.
- The first product feature which has been specified, ensures that a brief and concise description of the lecture is provided. **Safety** measures have to be taken when it comes to generating these summaries. There is always a high chance that the discussions that take place in class might bring about certain students' names or other personal details. It is highly essential for the product to ensure that no other personal details by any means are specified in the summaries which are being generated. Since this might lead to a lot of **privacy** issues, appropriate safety measures must be taken to ensure the privacy of the students as well as the teachers.
- We have identified that the main users of our product would be students, teachers, lecturers and other staff pertaining to academics. When it comes to accessing lecture videos, users have to be **authorized** by the university/institution and must have the required permissions to view these videos.
- In the same manner, the summary and the topics generated by our model must also be accessible only by the users who have been permitted by the respective institution. If the product is going to be integrated with the school's official website, the identity **authentication** utilized by the website would suffice. If the product is going to be stand alone, users will be authenticated by the product's own confirmation feature.
- **Availability** can also be an issue if the dataset is too huge to be handled by the local machine. In that case, cloud environments need to be considered.

6.4 High Level System Design

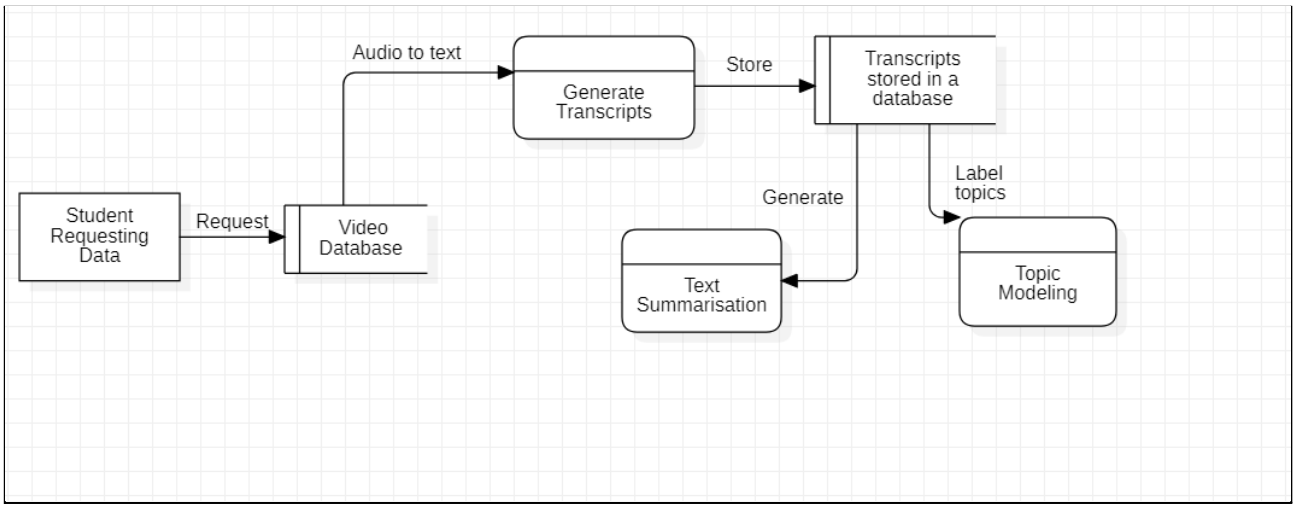

Fig 6.4.1: Logical Data Flow Diagram

The implementation of our project can be summarized as follows :-

I. We initially make use of the IBM watson API in order to extract the speech transcripts from the lecture videos.

II. We make use of TextRank, which is an extractive and unsupervised text summarisation approach in order to perform text summarization on the transcripts obtained.

III. We finally make use of sophisticated topic modelling techniques in order to generate labels for each lecture video and description.

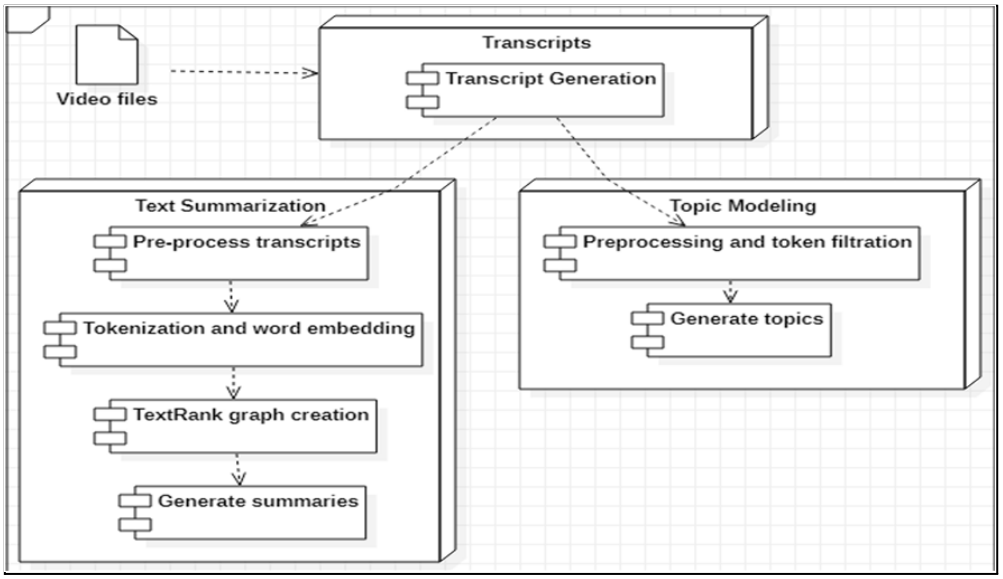

Fig 6.4.2 : Deployment Diagram

6.5 Design Description

6.5.1 Master Class Diagram

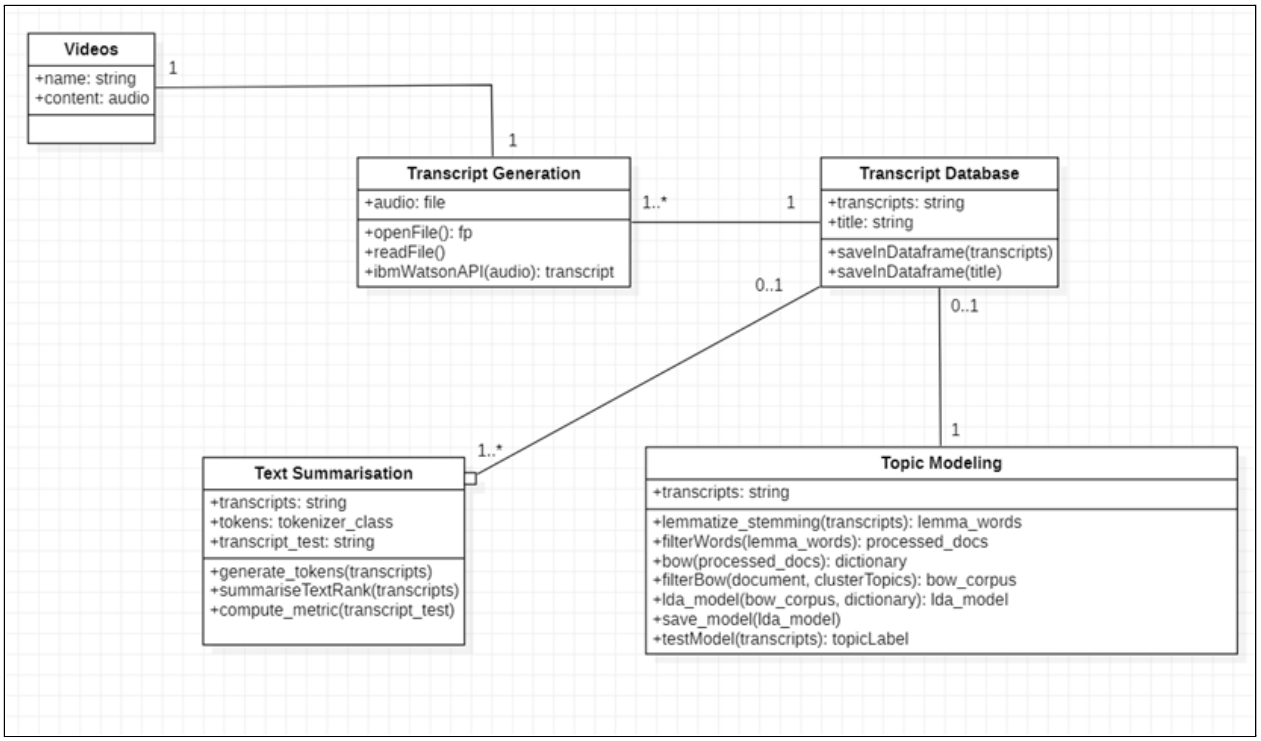

Fig 6.5.1 : Master Class Diagram

6.5.2 Reusability Considerations

The API, IBM watson, is reused to generate text transcripts. The modules like the TfidfTransformer, PunktSentenceTokenizer, CountVectorizer, etc are used for tokenization of transcripts and generation of summaries.

The transcripts can be downloaded to be reused for different functionalities like summarization and topic modeling.

Different functions written for concept extraction, filtering of concepts, lda model and labeling are reused for all the documents i.e. transcripts that are downloaded.

6.5.3 Data Flow Diagram

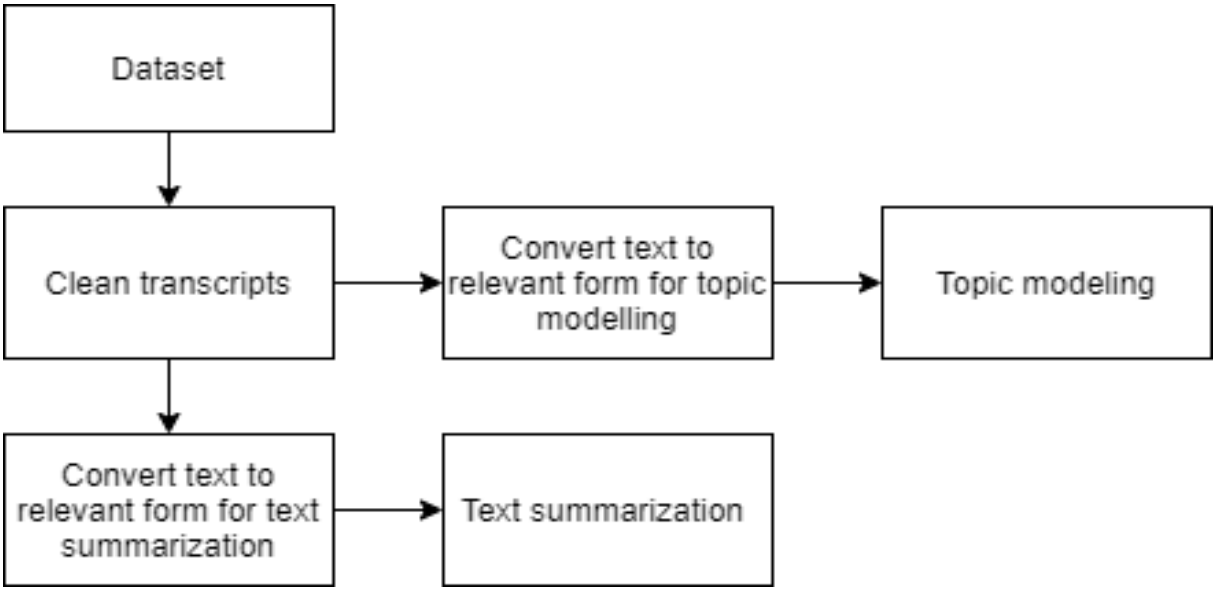

Fig 6.5.2 : Design approach in the form of a flow graph

6.5.4 Use Case diagram

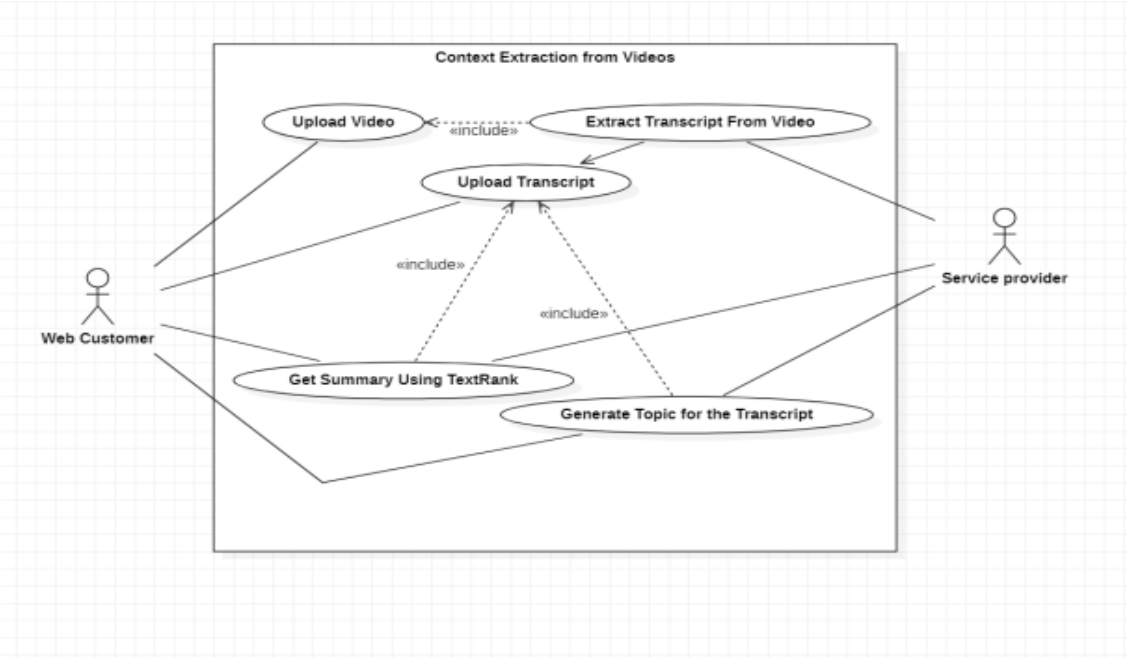

Fig 6.5.3 : Diagram depicting use cases which users would come across

6.5.5 Sequence Diagrams

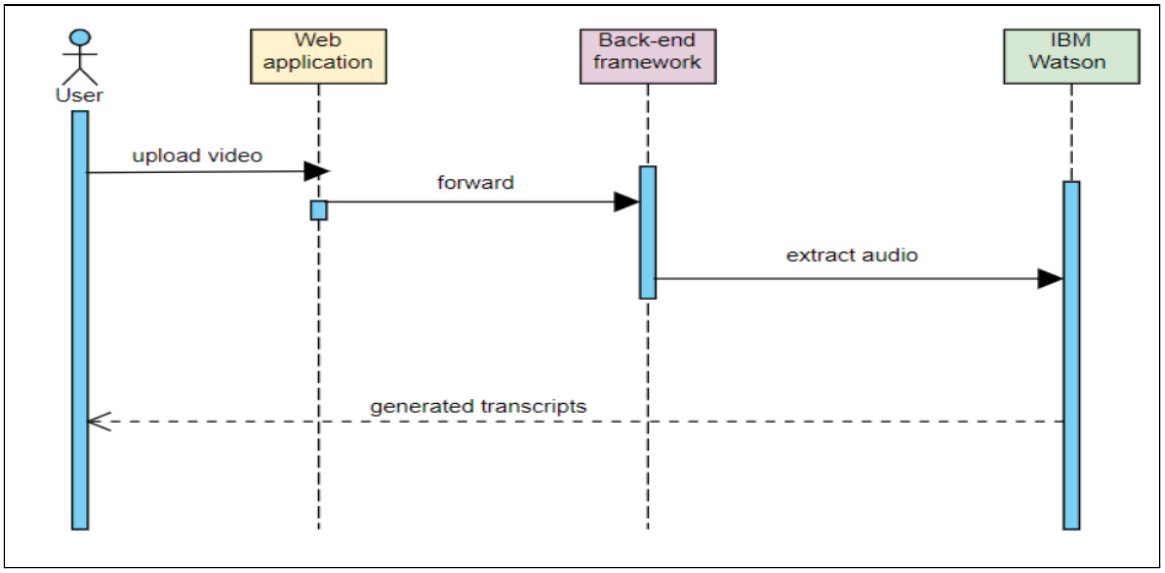

Fig 6.5.4 : Sequence Diagram for transcript generation

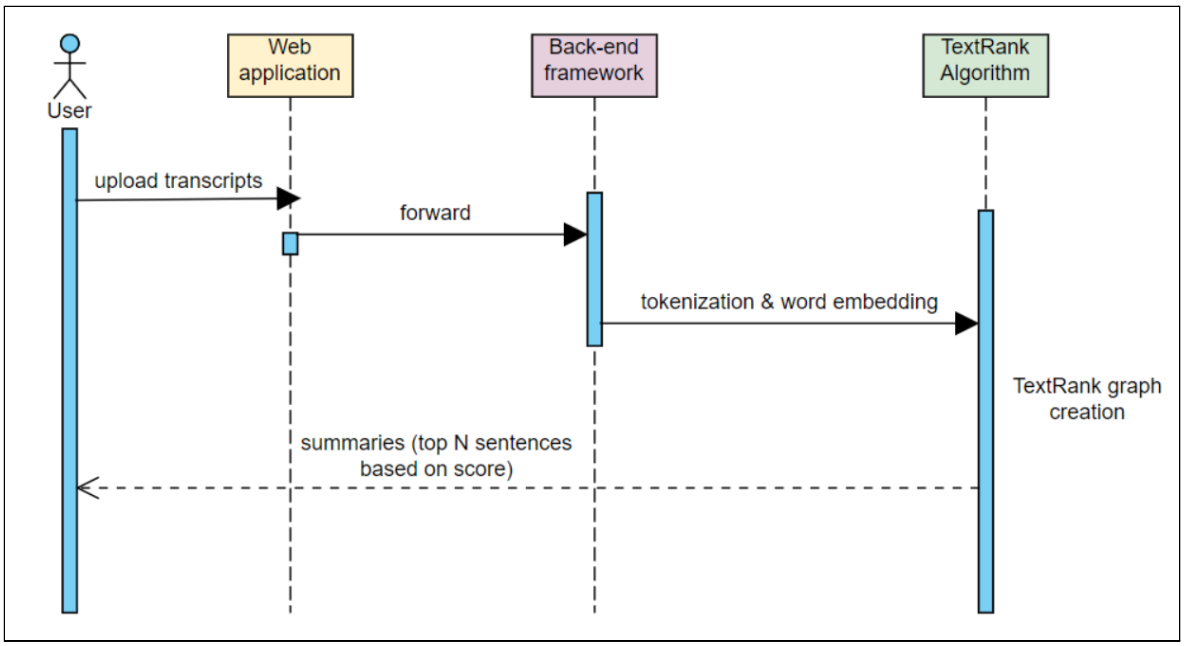

Fig 6.5.5 : Sequence Diagram for text summarisation

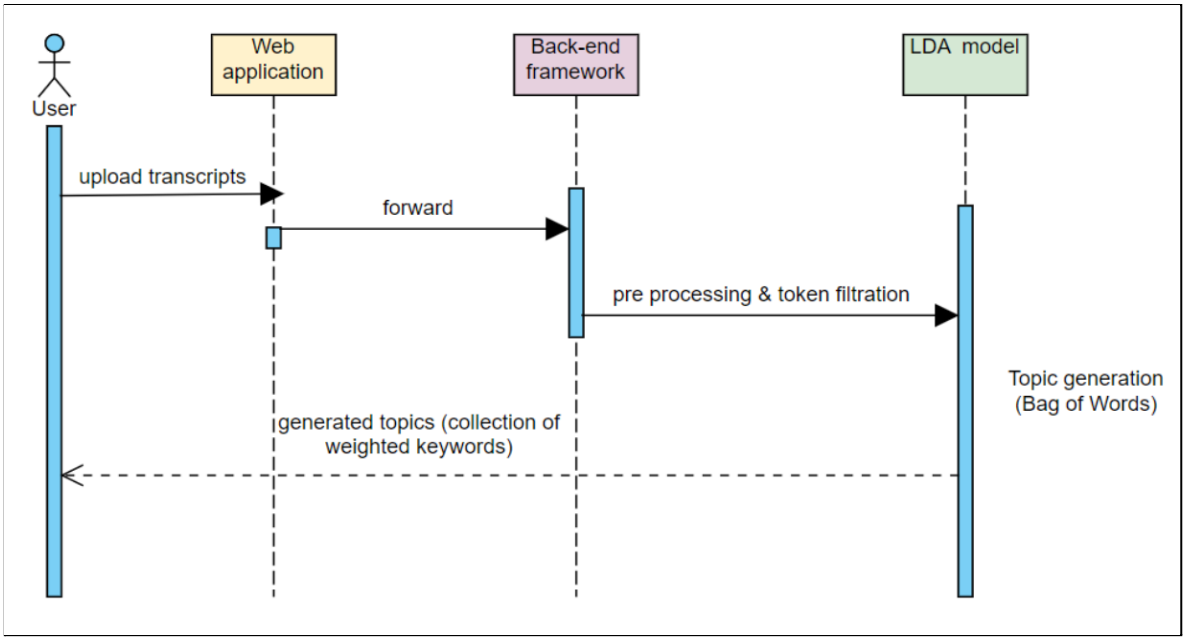

Fig 6.5.6 : Sequence Diagram for topic modeling

6.5.6 Report Layouts

Field	Data Type	Description	Is Mandatory
Input lecture video	MP4	A video to generate transcripts	Y

6.5.7 External Interfaces diagram

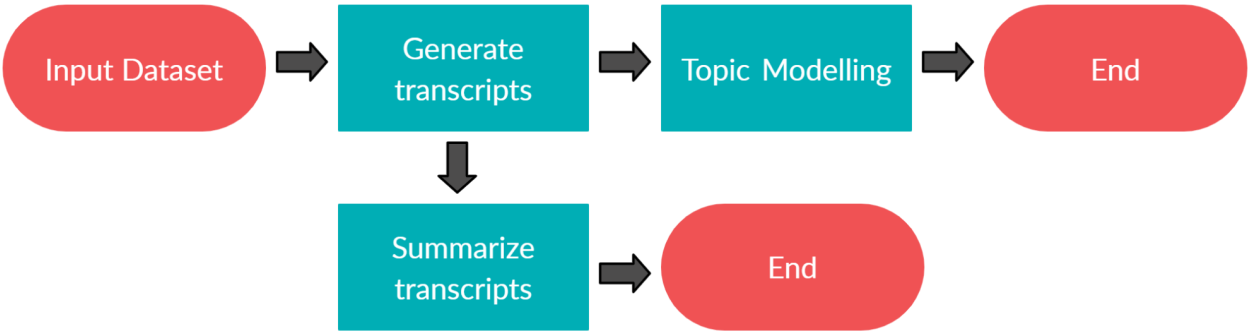

Fig 6.5.7 : External interfaces

6.5.8 Help

The Project Requirements Specification document, High Level Design document, Low Level Design document and the final report of this project could be used to aid in the usage of the system.

6.6. Design Details

6.6.1 Novelty

Lecturers in universities experience a lot of hassle in recording online lectures, giving the videos a title, trying to figure out which concepts are taught in the video and finally place the video under its respective topic. There are two main challenges to this traditional & conventional approach. The more general challenge is the fact that since this entire process is carried out manually, there is always a chance for human prone errors to take place. The other challenge is more pertaining to lecture videos recorded in universities. Most professors might begin teaching a new concept after finishing a previous concept. When multiple concepts are taught during a single lecture yet the video is identified by a single title, students may face difficulties in clearly identifying which concepts were taught in a particular lecture. Our product solves both the above challenges by making use of the transcripts of a lecture video in order to figure out the concepts of a lecture video and finally place the video under its respective topical cluster. This is a completely novel technique that we have used in order to automate the process of categorizing videos.

6.6.2 Innovativeness

Coming to the innovative aspects of our product, we have used the concepts of document clustering and topic identification to effectively yet efficiently categorize our lecture videos. Although document clustering has been an integral part of academia, there are hardly any products in the market which make use of document clustering to group videos. Hence existing methods, which make use of document clustering to organize written documents, have been utilized to arrange videos.

6.6.3 Interoperability

Currently our product is made particularly for lecture videos. In the future with slight changes to the functionalities of the framework, we believe we can extend out to other domains making use of all videos in general. With respect to operability under different OSes, currently our product is developed to run on Windows and Linux.

6.6.4 Performance

We aim to achieve a performance higher than the conventional approach. Although training the model by a time intensive process, once the model is trained we believe that the performance would be good due to the use of state of the art techniques for text summarization as well as topic modelling.

6.6.5 Security

Security is highly essential for our product. Lecture videos must be uploaded only by the concerned professors. Students who have enrolled for a particular course must be able to view videos of the courses for which they have enrolled.

6.6.6 Reliability

System reliability comprises software as well as hardware reliability. We aim to provide a failure-free software and minimize as many defects as possible before the final release. Other defects which are identified after the release will be patched accordingly.

6.6.7 Maintainability

Maintenance of the software is given utmost importance. We aim to ensure that software components can be modified with ease to correct faults and improve performance. Attributes of the model can be changed with ease in order to test its working in adapted environments.

6.6.8 Portability

Our software is designed to run on Windows and Linux as of now. Basic hardware requirements satisfying a decent RAM and main memory would be sufficient for the software to run on the client's host machine.

CHAPTER-7
IMPLEMENTATION AND PSEUDOCODE

We have finished our implementation on the proposed methodology we were planning to use to solve our given problem statement. We identified some of the important technologies we used in our implementation. For the audio processing and speech recognition tasks we planned to use the IBM Watson API, developed by IBM Research. For the text summarization, we planned to use the TextRank framework. For the topic set identification model, we planned to use a type of statistical modeling for discovering abstract “topics” that occur in a collection of documents. Latent Dirichlet Allocation (LDA) is an example of a topic model and is used to classify text in a document to a particular topic.

Apart from these frameworks, we planned to use advanced python libraries for data processing, feature extraction, exploratory data analysis, natural language processing and topic set identification(unsupervised machine learning). We planned to use github/vscode for software development and git for version control. We planned to use vscode for basic debugging. Jupyter Notebook and Spyder may also be used in the later stages of development. We ensured a thorough round of testing in order to limit the risks associated with our product and ensure that bugs are kept at a minimal level. With regards to our actual implementation we have laid out the roadmap to begin our extensive work going forward. The following snippet is a brief pseudo code explaining our proposed approach.

7.1 Exploratory Data Analysis:

The number of videos under the topic NLP is the least and Linear Algebra has the most number of videos.

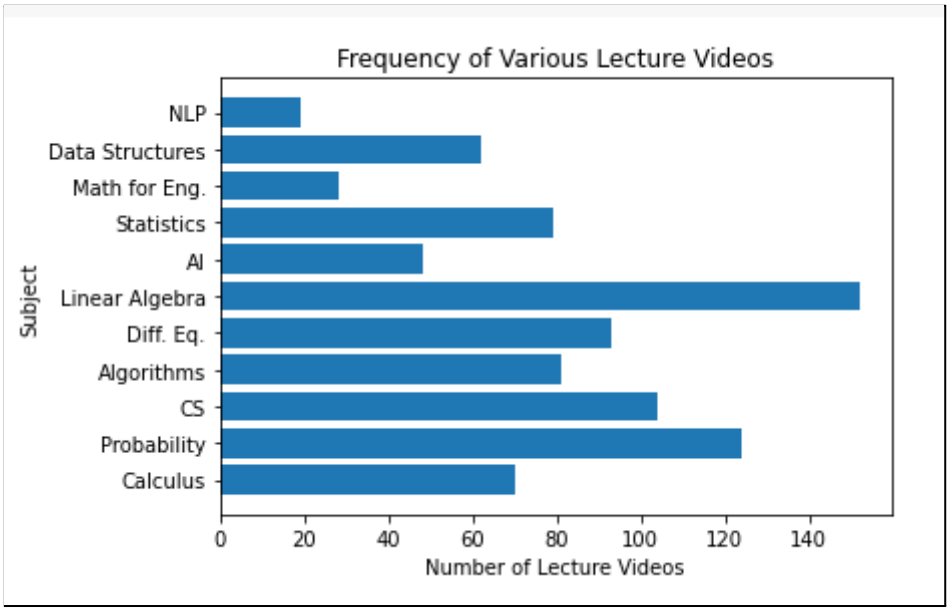

Fig 7.1.1 :Frequency of Various Lecture Videos

The longest transcript has around 15000 words and the shortest lecture video has around 100 words.

```
Longest Transcript has around 15293 number of words
Shortest Transcript has around 111 number of words
```

Fig 7.1.2: Number of words in the transcripts.

Words like **actually, well, OK, gt** (stopwords) appear more frequently than words like **random** and **variable**.

Fig 7.1.3: Frequency of words in the dataset

The dark shaded portion indicates that the topic is predicted correctly for the corresponding transcript; while the light area shows a lower recall score indicating wrong prediction of a topic.

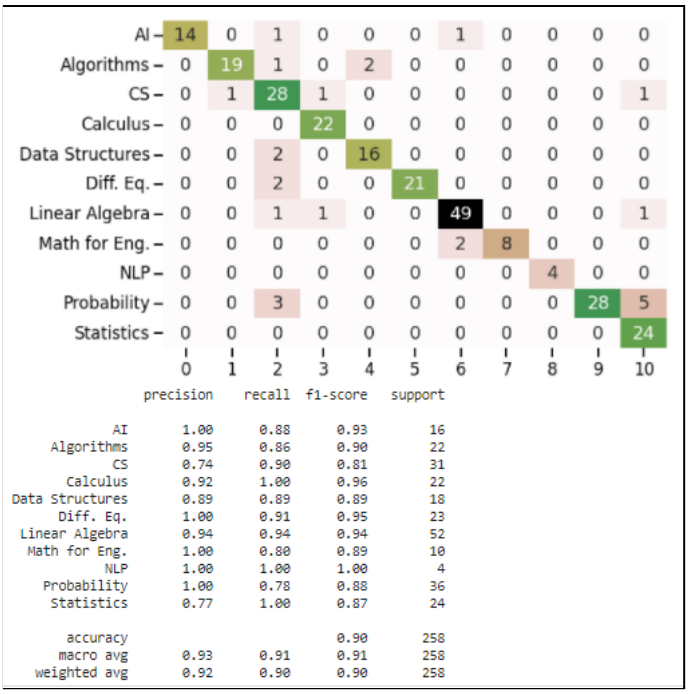

Fig 7.1.4:Confusion Matrix showing the actual and predicted topic details

7.2 Pseudocode:

```
// Pseudo Code explaining our methodology in a

// brief manner

START

    INPUT dataset comprising lecture videos
    EXTRACT relevant audio features from the videos
    APPLY speech recognition using the IBM Watson API
        to determine the transcripts

    SUMMARIZE the transcripts to get meaningful,
        syntactically and semantically correct
        brief descriptions using the TextRank
        framework

    IDENTIFY tokens from the transcripts/summaries
        relevant to the lesson taught in the lecture
        and filter out irrelvant tokens
        create a dictionary to filter out infrequent
        tokens

    IDENTIFY possible topics that can be used to
    categorize the lecture videos using LDA

    LABEL the given trancript using appropriate
    keywords

END
```

CHAPTER-8

RESULTS AND DISCUSSION

- All the functional and nonfunctional requirements proposed in the PRS have been implemented in our application.
- The three modules, namely the transcript generator, text summarizer & the topic model have been developed and tested thoroughly.
- The final evaluation metric of the text summarizer using TextRank gave us a BLANC score of 0.2 (approx).

	Unnamed: 0	Unnamed: 0.1	Cleaned_Transcript	Summary	Evaluation_Summary
0	0	0	The following content is provided under a reat...	And for now, will just to tell you that the ra...	0.199151
1	1	1	n this sequence of segments, we review some ma...	What we will talk about is sets, various defin...	0.340426
2	2	2	The following content is provided under a reat...	o you, as the programmer of this class, are go...	0.174203
3	3	3	The following content is provided under a reat...	ADENE: ADENE: ts a thats an n. PROFEOR: o we w...	0.208239
4	4	4	The following content is provided under a reat...	ADENE: n to the 1/8. ADENE: n to the 1/4. ADEN...	0.189708
...	...	...	...	...	...
85	85	626	Lets say that have some set V that is a subspa...	You could call this the image, because the tra...	0.237843
86	86	627	Lets do some more nonhomogeneous equations. o ...	Weve got the 1 e to the 4x plus 2e to the minu...	0.339181
87	87	628	The last time spent solving a system of equati...	The characteristic equation, then, is going to...	0.134228
88	88	629	The following content is provided under a reat...	And if we stop it at b, then the area were int...	0.119604
89	89	630	The following content is provided under a reat...	PROFEOR: o this thing in here is the answer to...	0.204393

Fig 8.1 :BLANC scores for few transcripts

- Similarly, the evaluation metrics obtained from the topic model using LDA gave us a Coherent Score of 0.37 for the Bag of Words word embedding technique and 0.32 for TF-IDF.

```
# Compute Coherence Score for LDA using BOW
coherence_model_lda = CoherenceModel(model=lda_model, texts=processed_docs, dictionary=dictionary, coherence='c_v')
coherence_lda = coherence_model_lda.get_coherence()
print('\nCoherence Score: ', coherence_lda)
```

```
Coherence Score:  0.3718279029107999
```

```
# Compute Coherence Score for LDA using TfIdf
coherence_model_lda = CoherenceModel(model=lda_model_tfidf, texts=processed_docs, dictionary=dictionary, coherence='c_v')
coherence_lda = coherence_model_lda.get_coherence()
print('\nCoherence Score: ', coherence_lda)
```

```
Coherence Score:  0.3134539600776823
```

Fig 8.2 :Coherence Scores for LDA using BOW and TF-IDF

- Based on the results, we can claim that our application has performed exceptionally well and can serve as a useful tool to procure concise description of lectures and figure out topics present in a particular video.

CHAPTER-9
CONCLUSION AND FUTURE WORK

Phase 2 concluded with us conducting an in-depth implementation of the project and all the relevant methodologies. In order to present the various functionalities of our application, we built a Flask Based web site. Any student/teacher can upload their lecture videos, get corresponding transcripts of the videos, choose to view a concise description of the video or even get appropriate topic labels with respect to a video. The summary & topic labels that are generated make use of the latest algorithms & machine learning techniques like TextRank and Latent Dirichlet Allocation.

When talking about the future, this project still has a wide range of possibilities to explore in detail. Its domain for example can be expanded to include topics in various fields outside of education in general such as sports and entertainment. Further enhancements and optimizations to the product could better refine its overall accuracy and performance in general.

REFERENCES/BIBLIOGRAPHY

[1] Domain taxonomy learning from text: The subsumption method versus hierarchical clustering

Publisher: Elsevier

Authors: Jeroen de Knijff, Flavius Frasincar, Frederik Hogenboom.

Date: 23 October 2012

[2] Survey on Abstractive Text Summarization

Publisher: IEEE

Authors: Nithin Raphal, Hemanta Duwarah, Philemon Daniel.

Date: 08 November 2018

[3] Semantic Text Summarization of Long Videos.

Publisher: IEEE

Authors: Shagan Sah, Sourabh Kulhare, Allison Gray, Subhashini Venugopalan,

Emily Prud'Hommeaux, Raymond Ptucha.

Date: 15 May 2017

[4] Multi-document Topic Segmentation Using Bayesian Estimation | IEEE Conference Publication

Publisher: IEEE

Authors: Pedro Mota; Maxine Eskenazi; Luísa Coheur

Date: 4-6 Feb. 2016

[5] SECTOR: A Neural Model for Coherent Topic Segmentation and Classification

Publisher: MIT

Authors: Sebastian Arnold, Rudolf Schneider, Philippe Cudré-Mauroux, Felix A. Gers and Alexander Löser.

Date: 2019

[6] Smart literature review: a practical topic modelling approach to exploratory literature review

Publisher: Journal of Big Data

Authors: Claus Boye Asmussen and Charles Møller

Date: 2019

[7] A Survey on Extractive Text Summarization

Publisher: ICCCSP

Authors: N.Moratanch ,S.Chitrakala

Date: 2017

[8] Deriving concept hierarchies from text

Publisher: UMASS

Authors: Mark Sanderson, Bruce Croft.

[9] Leveraging BERT for Extractive Text Summarization on Lectures

Author: Derek Miller

Date: 2019

[10] On How to Perform a Gold Standard Based Evaluation of Ontology Learning

Authors: Klaas Dellschaft and Steffen Staab

Date: 2006

[11] Guided HTM: Hierarchical Topic Model with Dirichlet Forest Priors

Publisher: IEEE

Authors: Su-Jin Shin, Il-Chul Moon

Date: 07 November 2016

[12] Text Tiling Text Segmentation Based on Hierarchical Dirichlet Process Model

Authors: Wei Ye, Xiaogang Gong, Quan Zhang and Xiaoming Ju

Date: 2019

[13] A Concept-Based Approach for Generating Better Topics for Web Search Results

Authors: N Mehala, Divyansh Bhatia

Date: 2020

[14] arvix.org

[15] https://www.geeksforgeeks.org/speech-to-text-using-ibm-watson-studio/

[16]https://www.analyticsvidhya.com/blog/2021/06/part-2-topic-modeling-and-latent-dirichlet-allocation-lda-using-gensim-and-sklearn/

APPENDIX A: DEFINITIONS, ACRONYMS AND ABBREVIATIONS

- **IBM Watson STT**: is a Speech to Text service provides an API to add speech transcription capabilities to applications. It combines information about language structure with the composition of the audio signal.
- **TextRank**: is a graph-based ranking model for text processing which can be used in order to find the most relevant sentences in text and also to find keywords
- **LDA:** is a generative statistical model that allows sets of observations to be explained by unobserved groups that explain why some parts of the data are similar.
- **GPU** (Graphical Processing Unit): A graphics processing unit (GPU) is a specialized electronic circuit designed to rapidly manipulate and alter memory to accelerate the creation of images in a frame buffer intended for output to a display device. GPUs are used in embedded systems, mobile phones, personal computers, workstations, and game consoles.
- **API** (Application Programming Interface): It is a software intermediary that allows two applications to talk to each other.
- **Bag of Words**: The bag-of-words model is a simplifying representation used in natural language processing and information retrieval. In this model, a text is represented as the bag of its words, disregarding grammar and even word order but keeping multiplicity.
- **TF-IDF**: Term Frequency — Inverse Document Frequency. This is a technique to quantify words in a set of documents. We generally compute a score for each word to signify its importance in the document and corpus. This method is a widely used technique in Information Retrieval and Text Mining.
- **Concept**: Concepts in Machine Learning can be thought of as a boolean-valued function defined over a large set of training data.
- **Topic**: Cluster of similar concepts is called a topic.

APPENDIX B: USER MANUAL

Our system has a simple user interface. Anyone interested in utilizing the service, will have to upload the lecture video in question and run the code, after which the system generates transcripts, topic models and a summarization of the transcripts.

APPENDIX C: RECORD OF CHANGE HISTORY

#	Date	Document Version No.	Change Description	Reason for Change
1.	25/9	3	TextRank instead of BERT for summarization	BERT Model not suitable for transcripts having greater than 1500 words
2.	25/9	3	Use of LDA	Use of LDA for the topic modelling approach
3.	25/9	3	Introduction of a Flask based website interface	Introduced GUI to improve client side experience
4.	25/9	3	Change of dataset	Decided to use a dataset having well punctuated transcripts and lectures pertaining to CSE

APPENDIX D: TRACEABILITY MATRIX

Project Requirement Specification Reference Section No. and Name.	DESIGN / HLD Reference Section No. and Name.	LLD Reference Section No. Name
4 Functional Requirements	3.1 Design goals	3.3 Text Summarizer 3.4 Topic Model 3.5 Flask Module
6 Non functional requirements	12 Design Details	2 Design Constraints, Assumptions & Details